Data Flow Diagram

Date	30 June 2026
Team ID	XXXXXX
Project Name	Emotion Detection and Learning Support Engine
Maximum Marks	2 Marks

Data Flow Diagram

The Data Flow Diagram (DFD) illustrates how information moves through the Voice-Based Concept Understanding Analyzer. It represents the interaction between external entities, system processes, data stores, and the flow of information from voice input to report generation. The diagram provides a high-level overview of how audio data is processed, analyzed, evaluated, stored, and finally presented to the user.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded Shape	External Entity	A person, organization, or external system that sends or receives information from the Voice-Based Concept Understanding Analyzer.
Rectangle with Number	Process	Represents a processing activity such as Speech-to-Text, Semantic Analysis, Audio Feature Extraction, or Report Generation.

Rectangle (Data Store)	Data Store	Stores information such as uploaded audio files and generated reports for future retrieval.
Labeled Arrow	Data Flow	Indicates the movement of information between external entities, processes, and data stores.

Data Flow Diagram Visualization

Student

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Input (Face / Voice / Text)

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Input Processing Module

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Emotion Detection Engine

(CNN / LSTM / Transformer)

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Emotion Classification

(Happy, Sad, Angry, Neutral, etc.)

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├──────────────────────► **Student Progress Database**

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Learning Support Engine

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Recommendation Engine

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Report Generator

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Dashboard & Personalized Learning Report

Description

The workflow begins when the **student** opens the application and provides input through a **webcam, voice recording, text input, or uploaded image**. The **Input Processing Module** preprocesses the captured data by cleaning text, extracting speech features, or detecting facial landmarks.

The processed input is then sent to the **Emotion Detection Engine**, where AI models such as **CNN**, **LSTM**, or **Transformer** analyze facial expressions, speech patterns, or textual sentiment to identify the user's emotional state (e.g., Happy, Sad, Angry, Neutral, Confused, or Stressed).

performance score. The Report Generator then creates a PDF report, stores it in the Reports Database, and displays the final evaluation results to the user.